

07	D
----	---

08	c
----	---

09	c
----	---

10	B
----	---

Question	Marking guidance	Additional comments/guidelines	Mark
01.1	Multiplies mass (either 2.40 kg or 2.98 kg) by 9.81 OR subtracts weight from upthrust ✓ 0.066(2) (N) ✓	Use of 9.8 for g gets 1 max Both marks can be given if working is clear but correct answer is rounded to 2sf or 1sf	2
01.2	MAX 2 Upthrust will change as balloon expands or the density of the atmosphere decreases ✓ air resistance which will change as the diameter of balloon changes ✓ air resistance will change as speed of balloon changes ✓	Allow references to wind applying a force (to change speed) condone upthrust explained e.g. "force provided by helium" Ignore references to temperature Allow 'air resistance will change as air density changes' owtte	2
01.3	Use of 75 for ΔT ✓ $n = 145 \text{ mol}$ ✓ Uses $\frac{3}{2}nRT$ or equivalent to get $-1.35 \times 10^5 \text{ (J)}$ ✓	Evidence may for MP1 may be seen in equation Condone lack of negative sign	3
01.4	Helium does work in expanding ✓ heat transfer is from balloon to surrounding because balloon is at a higher temperature ✓ Both Q and W are negative/ reference to $\Delta U = \Delta Q + \Delta W$ so internal energy falls ✓		3

Question	Marking guidance	Additional comments/guidelines	Mark
01.5	Refers to rate of energy transfer = $\frac{kA\Delta\theta}{L}$ ✓ Leading to ... Surface area increases (as balloon ascends) OR Thickness (of latex skin) decreases (as balloon ascends) ✓	Condone reference of $UA\Delta\theta$ for MP1 unless thickness argument used. Accept increase in temperature difference between inside and outside of the balloon for MP2	2
Total			12

Question	Marking guidance	Additional comments/guidelines	Mark
----------	------------------	--------------------------------	------

Question	Marking guidance	Additional comments/guidelines	Mark
02.1	Mentions latent heat of vaporisation (of working substance) ✓ Temperature (of working substance) falls OR is lower than the temperature of the contents ✓ Heat transfer because there is a temperature difference ✓		3
02.2	Uses average temperature difference of between 8 K and 12 K ✓ area $(3.0) \text{ m}^2$ ✓ Uses rate of heat transfer = $UA\Delta\theta$ to give an answer between 16 (W) and 24 (W) ✓	Expect to see 3.0 m^2	3
02.3	Uses $Q = mc\Delta\theta$ ✓ Divides by 4×3600 to give $28(.2)(\text{W})$ ✓	Expect to see $4.06 \times 10^5 \text{ (J)}$	2
02.4	Adds candidate's 02.3 + candidate's 02.2 + 60 ✓ Answer between 105 (W) and 111 (W) ✓		2
02.5	Energy transfer (to interior of house) is work done by pump plus energy extracted from ground ✓ Correct because more useful energy for same energy input ✓		2
Total			12

Question	Marking guidance	Additional comments/guidelines	Mark
----------	------------------	--------------------------------	------

6	B
---	---

7	B
---	---

8	B
---	---

9	C
---	---

10	A
----	---

11	D
----	---

12	D
----	---

14	B
----	---

8	B	2.0 10 1.0	AO2
---	----------	------------------	-----

13	c	$1500 \text{ J kg}^{-1} \text{ K}^{-1}$	AO2
----	----------	---	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Evidence of determination of area of floor OR determines difference in temperature ✓ Evidence of substitution into thermal conductivity equation ✓ 0.17 ✓ $\text{W m}^{-1} \text{K}^{-1}$ ✓	Expect to see $(A = 2.5 \times 4.2) = 10.5 \text{ (m}^2\text{)}$ Expect to see $(\Delta\theta = 45 - 22 =) 23 \text{ (K)}$ Condone missing or incorrect units in MP1 and MP2 E.g. for MP3: $1200 = \frac{k \times \text{their } A \times \text{their } \Delta\theta}{3.5(\times 10^{-2})}$ Allow POT error in substitution Allow substitution of incorrect thickness (4.2 or 2.5) An answer where the substitution of Boltzmann's constant for k is seen can only access MP1. Condone $\text{kg m s}^{-3} \text{K}^{-1}$ but do not allow k for K.	4	$3 \times \text{AO2}$ $1 \times \text{AO3}$

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Substitution into U-value equation With correct temperature difference ✓ 76 (°C)✓	Rate of energy transfer = $UA\Delta\theta$ Expect to see ($\theta_c - 22$) Calculator value is 76.4217687	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Idea that the (rate of) energy transfer depends on the temperature (of the heating system) ₁ ✓ With carpet, the temperature (of the heating system) is greater therefore rate of (wasted) energy transfer (to ground etc.) is greater ₁ ✓ ₂ ✓	Allow ecf for their 03.2 (compare with 45 °C) The first statement only gets MP1. MP1 and MP2 can be given for the second statement.	2	2 × AO4
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

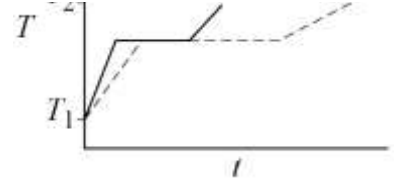
8	C	16 kJ kg^{-1}	AO3
---	---	-------------------------	-----

9	B	200 J	AO1
---	---	-------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Uses $A = 4\pi r^2$ ✓ Uses power = $\frac{kA\Delta\theta}{L}$ ✓ 32 to 34 ✓	Allow 0.02864 - 0.02736 for the radius. Expect $A = 9.85 \times 10^{-3} \text{ (m}^2\text{)}$ Expect 20.3 (W) Condone use of % sign in the answer	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Uses $Q = mc\Delta\theta$ with their m ✓ Uses $m = 4\pi r^2 t \rho$ OR $m = \frac{4}{3}\pi\rho(r_2^3 - r_1^3)$ ✓ 1460 (J) ecf from 02.1 ✓	Only penalise mixing up d and r once in 2.1 and 2.2 if use of d is seen clearly in working. Condone working out mass with volume of 1 sphere for mp2 Condone ecf for wrong area with $m = At\rho$ For mp3 allow 1436 OR 1470 OR 1403 (J) ecf for using d 5812 OR 5879 OR 5745 (J)	3	1 × AO1 2 × AO2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

08	B		AO3
----	---	---	-----

09	C	ignoring energy losses from the water heater	AO4
----	---	--	-----

10	C	window, door, wall	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Idea that: it allows for long duration of (close together) recordings OR the temperature change occurs too rapidly to measure this change using a thermometer ✓	Do not allow idea of convenience of placing sensor in test tube	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Idea that at this temperature latent heat is released (as the liquid changes to solid) ✓ Idea that latent heat 'released' during phase change is equal to energy lost to surroundings so temperature is constant ✓	Alternative for MP2: discussion around kinetic energy of particles being constant as potential energy changes	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Any three from: ✓✓✓ • calculation of a $\Delta\theta$ • use of $Q = mc\Delta\theta$ with any $\Delta\theta$ and either c • use of $Q = mL$ • their $Q = mc\Delta\theta$ for liquid added to their mL and their $E = mc\Delta\theta$ for solid Answer that rounds to 2.61×10^4 (J) ✓	eg $80^\circ\text{C} - 58^\circ\text{C}$ OR 22°C seen	4	$2 \times \text{AO2}$ $2 \times \text{AO3}$

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Evidence of use of a tangent drawn on Figure 1 to determine gradient at 300 s ✓ Answer in the range 0.15 to 0.17 (K s^{-1}) ✓		2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	Any two from: ✓✓ - determines rate of energy transfer from S $E = mc \times 0.055$ - uses their rate of energy transfer = $\frac{kA\Delta\theta}{L}$ to determine their $\Delta\theta$ - uses their reading at 400 s – their $\Delta\theta$ 29 (°C) ✓		3	AO2
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

12	B	$U_1 + Q - U_0$	AO1
----	---	-----------------	-----

14	C	$\text{kg s}^{-3} \text{K}^{-1}$	AO1
----	---	----------------------------------	-----

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Kinetic energy does not change ✓ Potential energy increases ✓		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Any two from: ✓✓ • uses $mc\Delta\theta$ for either copper or water • relates 36 kJ to sum of their $mc\Delta\theta$ • finds their $\Delta\theta$ and subtracts from 45 °C 15 (°C) ✓	Calculator value = 14.61384258	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Uses $mc\Delta\theta$ for 0.015 kg of water with $\Delta\theta = 55$ °C OR Relates 36 kJ to sum of their $mc\Delta\theta$ and mL ✓ 2.2×10^6 (J kg ⁻¹) ✓	If no other mark given, allow 1 mark for equating 36 kJ to $0.015 \times L$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Two suitable changes eg Use a lid on the can with reason eg Reduces heat loss (to the surroundings)✓ Use more water Reduces percentage uncertainty✓	Change and reason needed for a mark. Alternatives: • lid to prevent evaporative losses • deeper water to ensure none of the steam escapes • wait until water is boiling strongly to ensure steam is dry • insulate the tube/tube with lower heat capacity to ensure steam stays dry • stir the water before reading temp to ensure all water is at this temp • use a temperature range such that the initial temperature is the same number of degrees below and above the room temperature to ensure that energy transfers to and from the surroundings are equal.	2	AO4
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
----------	---------	--------------------------------	------	----

Unknown Unknown - Unit 4 - Q02

Question	Marking guidance	Mark	Comments
02.1	Use of $T = I\alpha$ ✓ Use of $\omega = \alpha t$ ✓ 0.219 (rad s ⁻¹) to at least 3 sf ✓	3	Use of equation is rearrangement or substitution
02.2	Use of $E = \frac{1}{2}I\omega^2$ leading to 1.2 x 10 ⁹ (J) to 1.3 x 10 ⁹ (J) ✓	1	